DISTIL: Demonstration Distillation for Sample-Efficient Imitation Learning

Anonymous submission

Abstract

Expert demonstrations are the dominant cost of imitation learning, yet interactive methods spend them reflexively: a per-state gate fires on the first uncertain step of a rollout and the expert corrects that miss, whatever it is. Under a fixed budget, what matters is *which* failures get corrected and *how* each corrective demonstration is placed. We present **DISTIL**, an interactive imitation-learning loop that distills the pool of possible corrections down to the few most informative demonstrations and spends the fixed budget only on those. DISTIL collects the current policy's failed rollouts, locates each failure's peak-loss step with the learner's own per-step training loss, describes it with a vision-language model, and assigns a root cause grounded in a task knowledge graph. The failures are clustered into recurring modes under a cross-round memory, and a reasoning LLM prescribes the round's one corrective demonstration behind a feasibility check with re-prescription. The learner is any policy f_θ with a per-step loss: we instantiate pure CNN and MLP classifiers on a GridWorld task and state- and image-based diffusion policies on four robot manipulation tasks. Under an identical 20-demonstration budget against six interactive baselines (five per setting), DISTIL attains the highest mean final success rate in all ten task–modality settings (e.g., 96.1% vs. 90.7% for Diff-DAgger on state Push-T; 95.3% vs. 89.6% on image Wipe), collects demonstrations with the highest per-demonstration information gain in every setting, and reports prescription confidences that predict realized policy improvement (Pearson $r = 0.82$–0.89).

Introduction

Behavior cloning regresses an expert's actions on the states the expert visits (Pomerleau 1988), and its well-known failure mode is covariate shift: the learner's own actions steer it off the demonstrated support, where errors compound over the horizon (Ross, Gordon, and Bagnell 2011). Interactive imitation learning fixes the distribution by aggregating expert corrections on states the *learner* visits (Ross, Gordon, and Bagnell 2011; Celemin et al. 2022), but the fix is paid for in expert effort: every correction is a demonstration an expert must produce. The query-efficient DAgger family therefore gates the expert behind a per-state predicate on action discrepancy, predictive uncertainty, novelty and risk, or the diffusion policy's own training loss (Zhang and Cho 2017; Menda, Driggs-Campbell, and Kochenderfer 2017, 2019; Hoque et al. 2021a; Lee, Kang, and Kuo 2025). All of these decide *when* to interrupt a rollout; the expert then corrects whatever failure happened to trip the gate.

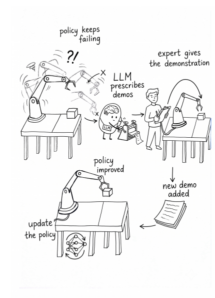

Figure 1: **DISTIL in a nutshell.** An LLM diagnoses the learner's failures and *prescribes* the one corrective demonstration the expert should provide; the demonstration is added and the policy retrained. Placement is part of the prescription: a demonstration can correct one recorded failure in place or start between several related ones.

This is wasteful under the constraint that binds in practice: a *fixed demonstration budget*. When only twenty corrective demonstrations can be collected, the first uncertain step of the current rollout is rarely the best place to spend one. Failures repeat: a when-to-query gate can correct the same recurring mode repeatedly while rarer modes go uncorrected for many rounds. Placement matters too: one demonstration started *between* several related failures can cover all of them, while an on-policy correction covers only its own rollout. These two decisions, the choice of failure mode and the placement of each demonstration, set what a fixed budget buys.

DISTIL (demonstration distillation) answers both questions with one loop (Figures 1 and 2). Rollouts of the current policy are collected, and the learner's own per-step training loss locates each failed episode's peak-loss step. A vision-language model (VLM) describes each failure from its start, highest-loss ($t^\star$), and end frames, and a reasoning LLM assigns a root cause grounded in a task knowledge graph. The failures are clustered into k recurring modes; a cluster memory carried across rounds rotates attention away from modes already corrected. A language model then *prescribes* the round's single corrective demonstration: a targeted on-policy correction of one recorded failure, or a bridging demonstration placed at a new start state covering several related failures. A feasibility check with re-prescription and a deterministic fallback vets every prescription, so no budget is spent on an infeasible prescription. The expert supplies exactly the prescribed demonstration, it is added to the dataset, and the policy is updated; *which* mode to correct and *where* to start are the two arms of the hybrid prescription. The name is the mechanism: out of the pool of every corrective demonstration the expert could provide, DISTIL distills the few most informative ones, the purest corrections, and the fixed budget is spent only on those.

DISTIL is policy-agnostic: it needs only a learner f_θ whose per-step training loss ℓ_t can be evaluated along rollouts (cross-entropy for the GridWorld CNN/MLP classifiers; denoising loss for the diffusion policies (Chi et al. 2023) on the robot tasks); the same loop runs unchanged over both families.

Contributions. **(1)** We reframe budget-constrained interactive imitation learning as demonstration distillation: decide *which* failure modes to correct and *how* to place each corrective demonstration, rather than only *when* to interrupt a rollout. **(2)** DISTIL, a policy-agnostic loop combining uncertainty flagging, VLM failure perception, knowledge-grounded root-cause reasoning, failure-mode clustering with cross-round cluster memory, and LLM demonstration prescription with a feasibility-checked re-prescription loop. **(3)** A controlled study on five tasks $\times$ two modalities against six baselines (five per setting): DISTIL has the best mean in all ten settings at the 20-demonstration budget, the highest per-demonstration information gain everywhere, and exposes a confidence signal that predicts realized improvement (Pearson $r = 0.82$–0.89). **(4)** A bridge between online and offline imitation learning: each round's feasibility-checked prescription is a standing artifact, so the loop can pause after its on-policy analysis and the expert can record the prescribed demonstration days later; expert time is decoupled from policy training time round by round.

Related Work

Interactive Imitation Learning

Behavior cloning dates to ALVINN (Pomerleau 1988). DAgger aggregates expert labels on learner-visited states as no-regret online learning (Ross, Gordon, and Bagnell 2011); Celemin et al. (2022) survey the field. Because vanilla DAgger queries constantly, query-efficient variants gate *when* to defer: SafeDAgger learns a discrepancy classifier (Zhang and Cho 2017); DropoutDAgger and EnsembleDAgger threshold dropout and ensemble uncertainty (Menda, Driggs-Campbell, and Kochenderfer 2017, 2019); HG-DAgger and LazyDAgger manage human-gated control switching (Kelly et al. 2019; Hoque et al. 2021b); ThriftyDAgger budgets interventions by novelty and risk (Hoque et al. 2021a); IWR and Sirius reweight samples around human takeovers (Mandlekar et al. 2020; Liu et al. 2023). Diffusion policies (Chi et al. 2023; Ho, Jain, and Abbeel 2020) capture the multimodal actions that make agreement-based uncertainty brittle, and Diff-DAgger queries on the diffusion training loss itself (Lee, Kang, and Kuo 2025). All decide when to interrupt. DISTIL adds the round-level decisions they leave open: it reasons over the set of a policy's failures, keeps a memory of what was corrected, and places each demonstration's start.

Foundation Models for Failure Reasoning

Language and vision-language models steer robot data collection and control: SayCan grounds plans in affordances (Ahn et al. 2022), Inner Monologue closes a textual feedback loop (Huang et al. 2022). A second line emits executable structure: policy code (Liang et al. 2023), 3D value maps (Huang et al. 2023), or reward functions (Ma et al. 2024). Closest to us is failure reasoning: REFLECT narrates and corrects robot failures (Liu, Bahety, and Song 2023), and AHA reasons over manipulation failures (Duan et al. 2025). These systems reason about one trajectory at a time; DISTIL aggregates VLM-perceived failures across rollouts, clusters them into modes, and narrows the LLM's output to one auditable artifact: a feasibility-checked prescription for a single corrective demonstration.

Active Selection, Representations, and Resets

Deciding which failures deserve a demonstration is batch active learning over failures: pool-based selection under a budget (Settles 2009), core-set/k-center coverage (Sener and Savarese 2018), and farthest-point sampling (Eldar et al. 1997). Whereas these select unlabeled inputs, DISTIL selects failure *modes* and adds a cross-round memory so coverage accumulates over the whole run. Bridging demonstrations placed at prescribed start states relate to reverse-curriculum reset generation (Florensa et al. 2017); DISTIL's start states follow clustered failure evidence rather than a difficulty schedule.

Method

Setup: Budgeted Interactive Imitation

Each task is a finite-horizon decision process with states s, actions a (discrete moves on GridWorld; relative joint-position deltas on the robot tasks), and an expert $\pi^\star$ (a human on GridWorld; privileged controllers on the robot tasks). The learner is *any* policy f_θ trained on an aggregated demonstration dataset $\mathcal{D}$ by minimizing a per-step loss ℓ,

$$\theta^\star = \arg\min_\theta \; \mathbb{E}_{(s,a)\sim\mathcal{D}}\big[\,\ell(f_\theta; s, a)\,\big], \qquad (1)$$

where ℓ is the categorical cross-entropy over the expert's action labels for the GridWorld classifiers (pure CNN on images, MLP on states) and the denoising (v-prediction) loss for the diffusion policies on the robot tasks (Ho, Jain, and Abbeel 2020; Chi et al. 2023). Evaluated along a rollout at the learner's own executed actions, the loss yields a per-step uncertainty signal

$$\ell_t = \ell\big(f_\theta;\, s_t,\, a_t^{\mathrm{nov}}\big), \qquad (2)$$

where a_t^{nov} is the learner's own executed action: the training-loss functional evaluated at the executed action rather than

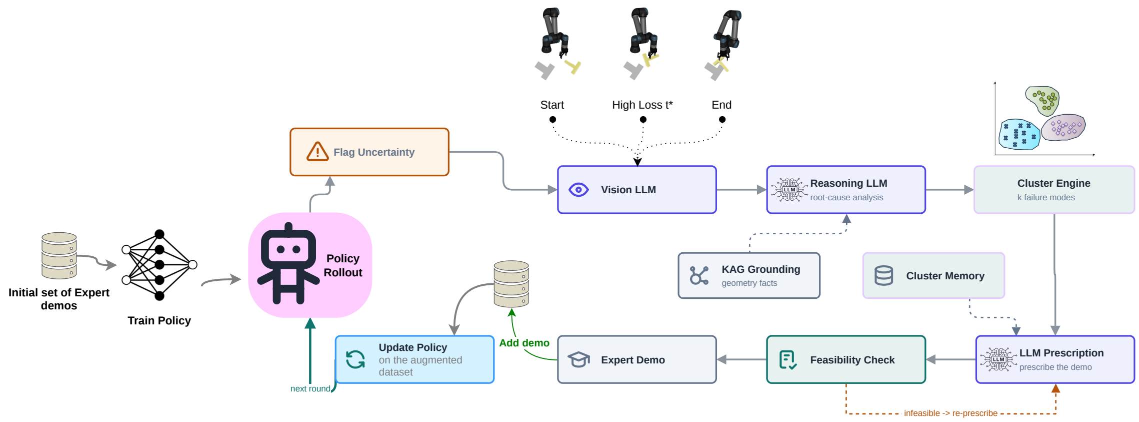

Figure 2: **The DISTIL loop.** Rollouts of f_θ are monitored for uncertainty via the per-step training loss; a VLM summarizes each flagged failure's start, highest-loss ($t^\star$), and end frames, and a reasoning LLM assigns root causes grounded in a task knowledge graph (knowledge-augmented generation, KAG); failures are clustered into k recurring modes with cross-round memory, the LLM prescribes the round's corrective demonstration behind a feasibility check with re-prescription, and the accepted demonstration augments the dataset before the policy is updated.

an expert label, a self-uncertainty signal scoring how far the visited pair sits from the training distribution (Lee, Kang, and Kuo 2025); the stochastic denoising loss is averaged over sampled noise and timestep draws. Interactive learning proceeds in rounds: round r produces one corrective demonstration d_r (a segment of expert state–action pairs), and

$$\mathcal{D}_{r+1} = \mathcal{D}_r \cup d_r, \qquad \theta_{r+1} = \arg\min_\theta \mathbb{E}_{\mathcal{D}_{r+1}}[\,\ell\,]. \tag{3}$$

All methods receive the same hard budget of B=20 corrective demonstrations, one per round; the only degree of freedom is how each demonstration is chosen. The dataset grows every round; the re-optimization of Eq. 3 runs from scratch at a per-task cadence shared by every method (every round on GridWorld, every fourth round on the robot tasks; protocol in Experiments).

Baseline Query Rules

Each uncertainty-gated baseline instantiates one template, a scalar score gated by a threshold; in its per-state form the expert takes over at the first crossing,

$$\mathrm{Query}_m(s_t) = \mathbf{1}\big[\,\mathrm{score}_m(s_t) > \tau_m\,\big]. \tag{4}$$

SafeDAgger scores the novice–expert action discrepancy $\|a_t^{\mathrm{nov}} - a_t^{\mathrm{exp}}\|_2$ (τ=0.1; on GridWorld, the fraction of rollout steps off the expert's action labels) (Zhang and Cho 2017). DropoutDAgger draws N=10 stochastic action samples and fires when the fraction landing within a τ-ball (τ=0.1) of the expert action drops below p=0.9 (Menda, Driggs-Campbell, and Kochenderfer 2017). Both rules deviate from their published forms (no learned safety classifier; expert-referenced instead of variance-only dropout), so every baseline's gate may consult the expert's action at any visited step for free; only recorded demonstration segments count against the budget. DISTIL's flagging never consults the expert, so the uncounted access favors the baselines. EnsembleDAgger fires on the doubt (variance) of M=5 independently trained members or on their mean discrepancy (χ=0.05, τ=0.1) (Menda, Driggs-Campbell, and Kochenderfer 2019). ThriftyDAgger fires on novelty or on learned task risk $1 - Q_\psi(s,a)$, with thresholds calibrated on the pool's score quantiles on GridWorld and a fixed conservative initialization on the robot tasks (Hoque et al. 2021a). Stagger, the round-level random control, fits no per-state template: each round it corrects one uniformly chosen recorded failure. On the robot tasks we also run Diff-DAgger (Lee, Kang, and Kuo 2025), whose score is the diffusion loss of Eq. 2 itself: with $\hat{F}$ the empirical CDF of training-set losses, K the patience window (K=1), and α=0.99,

$$\mathrm{Query}_{\mathrm{Diff}}(s_t) = \mathbf{1}\big[\,\hat{F}(\ell_u) > \alpha \;\; \forall u \in (t{-}K,\, t]\,\big], \tag{5}$$

recalibrated at every retraining. The rosters differ because Diff-DAgger's rule *is* the diffusion policy's own training loss: it requires a diffusion learner and does not apply to the GridWorld classifiers, where Stagger is the matched random control. On the robot tasks the control is Diff-DAgger; we did not run the random control there, so those cells lack a random-allocation control that would isolate clustering and memory from failure replay alone. The roster is six baselines overall, five per task. The template is applied in two forms. On the robot tasks the gate runs per state: the rollout continues until the first threshold crossing, bounded by an episode cap, and the expert corrects that rollout from the crossing step. On GridWorld each method scores every layout in the round's pool with its rule and corrects the highest-scoring layout whose gate fires; when no gate fires, the pool scoring falls back to the highest-scoring failure, so the round still yields one demonstration. Either way, each rule decides *when* to interrupt; none compares failures, remembers corrected modes, or chooses where the demonstration starts.

The DISTIL Loop

DISTIL replaces the per-state gate with a round-level analysis of the whole failure set, run identically for every learner. Algorithm 1 gives the loop.

Uncertainty flagging and failure perception. Each round rolls the current policy (a freshly sampled pool of 20 layouts on GridWorld; 60 episodes on the robot tasks) and collects the failure set $\mathcal{F}_r = \{f_1, \dots, f_N\}$: the episodes that fail the task. Along each failed episode the per-step loss ℓ_t (Eq. 2) is logged; an adaptive threshold (the 95th percentile of the episode's loss distribution) selects the steps stored as failure-point candidates, the episode's peak-loss step always among them, and each failure keeps its peak-loss step

$$t_i^\star = \arg\max_t \ell_t^{(i)}, \qquad \mathrm{peak}_i = \max_t \ell_t^{(i)}, \tag{6}$$

taken as the failure point, the argmax running over the stored candidates. The VLM (Qwen3-VL-32B; Qwen Team 2025b) then describes three frames per failure (start, $t_i^\star$, and end) and produces a textual failure report (Liu, Bahety, and Song 2023; Duan et al. 2025). The reasoning LLM (Qwen3-32B; Qwen Team 2025a) assigns each failure a root cause (e.g., wrong approach, no contact, wrong orientation, timeout) and a task phase; the VLM runs at low reasoning effort and the LLM at high, both capped at 16,384 output tokens. Both stages are grounded by knowledge-augmented generation (KAG): a per-task knowledge graph (embodiment, object and goal geometry, workspace bounds, failure taxonomy, per-mode prescription rules) rendered into the prompt.

Failure-mode clustering. Each failure is featurized by a descriptor anchored at $t_i^\star$. For state-based robot runs this is the 6-D quaternion-free geometric vector

$$\phi_i = \big[\, p_{i,x},\ p_{i,y},\ \sin\theta_i,\ \cos\theta_i,\ \rho_i,\ \delta_i \,\big], \tag{7}$$

with $(p_{i,x}, p_{i,y}, \theta_i)$ the object's planar pose, $\rho_i = t_i^\star / T_i$ the task progress, and δ_i the end-effector contact distance; for image-based runs the descriptor is a frozen R3M embedding (Nair et al. 2022) of the same three frames, PCA-reduced to $k' = \min(16, N{-}1)$ dimensions. On GridWorld, ϕ_i uses the agent's cell, its signed offset to the goal, progress, and Manhattan distance-to-goal. The standardized descriptors $\tilde{X}$ are clustered agglomeratively, with the number of modes chosen by maximum mean silhouette,

$$k^\star = \arg\max_{k \in [2,\, k_{\max}]} \mathrm{sil}(k), \tag{8}$$

with $k_{\max} = \max\big(2, \min(6, N{-}1)\big)$. With $N \le 3$ the sweep is skipped and each failure is its own cluster (the image-run PCA passes features through for $N \le 2$), so a lone failure is targeted directly; a rollout set with no failures yields no demonstration and no budget charge (the budget counts recorded demonstrations), and the loop rolls out again; re-rolls are bounded by the run's episode cap (on GridWorld, a cap on consecutive failure-free pools). Each cluster C keeps its representative $\mathrm{rep}(C) = \arg\min_{i \in C} \|\tilde{X}_i - \bar{X}_C\|_2$, centroid geometry computed on raw poses, and mean peak loss $\bar{L}_C$; the dominant cluster $C^\star$ is the largest, breaking ties by $\bar{L}_C$.

Cluster memory. Because the failure distribution shifts after every retraining, a persistent mode could absorb the entire budget. DISTIL therefore remembers previously corrected cluster centroids, penalizes recently covered regions with a recency-discounted Gaussian kernel, and rotates the target among clusters that rival the dominant one:

$$P_{\mathrm{mem}}(c) = \sum_{(r_i, c_i) \in \mathrm{Mem}} \gamma^{\,r - r_i}\ \exp\Big(-\tfrac{\|c_{xy} - c_{i,xy}\|_2^2}{2\sigma^2}\Big),$$
$$C_{\mathrm{tgt}} = \arg\max_{C:\, |C| \ge |C^\star| - 1} \big(\, \bar{L}_C - \lambda\, P_{\mathrm{mem}}(c_C) \,\big), \tag{9}$$

with $\gamma{=}0.6$, $\sigma{=}0.06$, $\lambda{=}1$, and c_C the centroid of cluster C. Centroids are stored in raw object-pose coordinates (meters on the robot tasks) whatever the clustering features, so the kernel is untouched by per-round standardization or the changing PCA dimension of image descriptors; σ is in those units. The kernel distance is planar (c_{xy}) only: stored centroids carry a yaw that never enters P_{mem}, so on Push-T coverage of one mode also discounts a differently oriented mode at the same xy. Because a centroid is appended to Mem at the end of its round, P_{mem} is $\gamma = 0.6$ at a centroid covered in the previous round and approaches 1 only as coverages accumulate, so $\lambda{=}1$ lets one recent coverage offset roughly half a unit of mean peak loss. GridWorld centroids live in grid-cell coordinates, where $\sigma{=}0.06$ is a small fraction of one cell: the kernel reduces to an identical-centroid check shaped by the recency discount γ alone. Lift and Door sit at the opposite extreme: against their narrow reset ranges (± 0.03 m; roughly ± 0.013 m) the kernel is nearly flat, so the memory acts there as a recency-weighted global discount rather than a spatially selective penalty. The same constants are used unchanged on every task and both loss families, at the price of these two degenerate regimes. The constraint $|C| \ge |C^\star| - 1$ keeps each round on near-dominant modes; rare modes are reached across rounds: corrected modes shrink after retraining and are memory-penalized, so a formerly rare mode rises to near-dominant size and takes a later round. The prescription is also conditioned on a small diverse context set S ($|S| \le \kappa{=}3$): the target representative is forced in, the worst-peak-loss failure is seeded, and remaining slots are filled by farthest-point selection in descriptor space (Sener and Savarese 2018; Eldar et al. 1997).

Demonstration prescription. Given the clustered evidence, root causes, and cited context failures, the LLM prescribes the round's single corrective demonstration together with a self-reported confidence. The prescription is a hybrid decision with two arms, both in play wherever the randomized quantity is an object pose (Wipe and GridWorld, below, are the exceptions). *Targeted correction*: the LLM selects one recorded failure; its scene is re-instantiated, the current policy replays to its divergence point $t^\star$, and the expert takes over from there. *Bridging placement*: the LLM instead places the demonstration at a new start state ξ positioned between two or three cited failures of the same mode, so one demonstration covers all of them. The bridge pose is anchored to the target representative A: the commanded offset is L_2-capped in position, clamped in orientation, and projected onto the

KAG workspace bounds $\mathcal{W}$,

$$\begin{aligned}
\Delta_{xy} &= \mathrm{cmd}.p_{xy} - p_{A,xy}, \quad \Delta_{xy} \leftarrow \Delta_{xy} \min\Big(1, \tfrac{\Delta_{\max}}{\|\Delta_{xy}\|_2}\Big), \\
\Delta_\theta &= \mathrm{clamp}\big(\mathrm{wrap}_\pi(\mathrm{cmd}.\theta - \theta_A),\ \pm\theta_{\max}\big), \\
\xi &= \mathrm{clamp}_{\mathcal{W}}\big(p_A + [\Delta_{xy}, 0],\ \theta_A + \Delta_\theta,\ q_A\big).
\end{aligned} \tag{10}$$

Equation 10 gives the Push-T mechanism, with $(\Delta_{\max}, \theta_{\max}) = (0.06\,\mathrm{m}, 0.4\,\mathrm{rad})$ and q_A the representative's remaining pose components (height and full orientation), carried over unchanged. Lift and Door replace the offset cap with an absolute clamp: the bridge start is the mean xy of the cited failures, clamped to the task's native reset range, so a prescribed start can never leave the reset distribution, and $\Delta_{\max}$ denotes the half-width of that clamp range about its center ($0.03\,\mathrm{m}$ in x and y on Lift; 0.0135 and $0.013\,\mathrm{m}$ on Door), with $\theta_{\max} = 0$ (Lift has no yaw randomization; the Door bridge reuses the representative failure's orientation and shifts xy only). On Wipe the randomized quantity is a whole dirt-marker path, so bridging is infeasible and every prescription is a targeted correction. On GridWorld the prescription is a corridor layout (start, goal, obstacles, cell sequence) that the human expert is constrained to follow.

Feasibility check and re-prescription. Every prescription is validated before any expert effort is spent: robot prescriptions must respect $\mathcal{W}$ and cite real recorded failures; GridWorld corridors must stay on the grid, avoid the obstacles, and connect start to goal under an $A^\star$/BFS path-validity check. An infeasible or empty prescription is rejected and re-prescribed with the rejection fed back; after at most five attempts the round falls back deterministically to a targeted correction of the nearest untried recorded failure, so a round always yields a valid demonstration. The accepted demonstration is added: for a targeted correction, $d_r = \{(s_t, \pi^\star(s_t)) : t \geq t^\star\}$, the expert segment from the divergence point; for a bridge, the full expert demonstration started at ξ. The target centroid is appended to the memory and the policy is updated (Eq. 3).

Bridging online and offline imitation. DAgger-family methods keep an expert on call throughout training; classical behavior cloning collects demonstrations offline, then trains. DISTIL bridges the two. Each round's prescription is a standing artifact: the loop can run its rollout, clustering, and prescription analysis on-policy and then pause indefinitely; the prescribed start remains valid, and the expert can return days later to record that round's demonstration. The decoupling is per round (the next round's analysis depends on the retrained policy), but at no point must the expert supervise a rollout: expert time reduces to scheduled recordings of prescribed demonstrations.

Algorithm 1 The DISTIL loop (one demonstration per round).

Input: initial demos $\mathcal{D}_0$, expert $\pi^\star$, budget B=20
Output: trained policy f_θ

```
train f_θ on D_0 (Eq. 1);  Mem ← ∅
for r = 1 to B do
    roll out f_θ; collect failed episodes; record t*_i, peak_i
    (Eqs. 2, 6); VLM describes start / t*_i / end; reasoning
    LLM assigns KAG-grounded root causes
    featurize (Eq. 7); cluster into k* modes (Eq. 8); rotate
    target via cluster memory (Eq. 9); build context set S
    LLM prescribes with confidence: targeted correction
    of a cited failure or bridging start ξ (Eq. 10); re-
    prescribe while infeasible; after bounded attempts fall
    back to the nearest untried failure
    expert provides d_r;  D_r ← D_{r−1} ∪ d_r; append target
    centroid to Mem; retrain f_θ on D_r (Eq. 3, at the shared
    per-task cadence)
end for
return f_θ
```

Experiments

We ask three questions. **(Q1)** Under an identical 20-demonstration budget, does DISTIL reach a higher final success rate than when-to-query baselines? **(Q2)** Does it collect more informative demonstrations, and how does information gain relate to allocation across failure modes? **(Q3)** Are the LLM's prescription confidences predictive of realized improvement, and are the discovered failure modes semantically meaningful?

Setup

Tasks and experts. **GridWorld** $5{\times}5$: the agent moves from a start cell to a goal cell with four discrete moves on a $5{\times}5$ grid containing three obstacles; an episode succeeds when the agent reaches the goal. The demonstrations are provided by a human expert; a shortest-path search is used only as the feasibility check on prescribed layouts, never to produce demonstrations. **Push-T** (ManiSkill3, Tao et al. 2024; task from Implicit BC, Florence et al. 2021): a Franka Panda with a stick end-effector pushes a T-shaped block onto a fixed goal pose; the expert is a privileged PPO policy. **Lift**, **Wipe**, and **Door** (robosuite, Zhu et al. 2020, evaluation following robomimic, Mandlekar et al. 2021): a UR5e arm lifts a cube off the table (Robotiq-85 parallel-jaw gripper), wipes a mark off the table (wiping-pad end-effector), and opens a door (Robotiq-85), each with a scripted motion-planner expert. Every task runs in a *state* modality (privileged proprioception and object state) and an *image* modality (RGB observations hiding object state).

Learners. On GridWorld the image learner is a pure CNN and the state learner an MLP (cross-entropy on the expert's action labels). On all four robot tasks every method shares one diffusion-policy backbone (Chi et al. 2023; Ho, Jain, and Abbeel 2020), a conditional 1-D U-Net denoiser with a v-prediction objective, paired with an MLP state encoder or an end-to-end finetuned R3M ResNet-18 image encoder with spatial-softmax keypoints (Nair et al. 2022).

Protocol and metrics. Every method starts from the same behavior-cloned initialization per seed (20 initial demonstrations, excluded from the budget) and then collects expert demonstrations under the same hard budget of **20**, one per round (Eq. 3). Each demonstration joins the dataset when

Table 1: **Final held-out success rate (%) at the 20-demonstration budget**: mean and one standard deviation over each cell's held-out evaluation records at the budget, pooled across seeds by the evaluation pipeline (not per-seed means and not one final fixed-set evaluation per seed; per-cell record counts vary and are not seeds $\times$ episodes); 9 seeds per GridWorld cell, 5 per robot cell. Columns abbreviate SafeDAgger, DropoutDAgger, EnsembleDAgger, ThriftyDAgger; Stagger runs on GridWorld, Diff-DAgger on the robot tasks. Best mean per row in **bold**: DISTIL is highest in all ten settings; spreads overlap the best baseline in most cells (see text).

Task	Obs.	Safe	Dropout	Ensemble	Thrifty	Stagger	Diff-DAgger	DISTIL (Ours)
GridWorld 5×5	state	85.3 ± 2.7	84.9 ± 2.5	86.2 ± 2.1	86.8 ± 2.0	85.7 ± 1.5	–	$\mathbf{89.9 \pm 1.3}$
GridWorld 5×5	image	86.1 ± 2.8	85.8 ± 2.6	85.7 ± 2.2	87.1 ± 1.9	86.6 ± 2.3	–	$\mathbf{89.6 \pm 1.8}$
Push-T	state	82.0 ± 6.8	84.8 ± 6.1	85.9 ± 5.8	83.2 ± 7.2	–	90.7 ± 4.5	$\mathbf{96.1 \pm 4.5}$
Push-T	image	78.1 ± 7.8	82.1 ± 6.9	83.2 ± 6.6	79.3 ± 8.1	–	89.0 ± 4.8	$\mathbf{93.9 \pm 4.9}$
Lift	state	99.2 ± 1.6	99.2 ± 1.0	99.2 ± 1.0	98.8 ± 2.4	–	99.2 ± 1.0	$\mathbf{100.0 \pm 0.0}$
Lift	image	99.6 ± 0.8	97.2 ± 3.5	98.8 ± 1.6	99.6 ± 0.8	–	99.6 ± 0.8	$\mathbf{100.0 \pm 0.0}$
Wipe	state	88.0 ± 2.5	89.6 ± 4.1	90.8 ± 4.3	90.0 ± 2.5	–	90.4 ± 6.0	$\mathbf{95.5 \pm 6.0}$
Wipe	image	69.6 ± 5.3	83.2 ± 6.8	84.4 ± 7.1	69.2 ± 9.0	–	89.6 ± 3.2	$\mathbf{95.3 \pm 3.2}$
Door	state	93.2 ± 5.2	92.8 ± 2.7	88.8 ± 7.0	89.6 ± 3.9	–	95.2 ± 4.3	$\mathbf{98.4 \pm 4.2}$
Door	image	92.4 ± 3.2	88.8 ± 3.3	86.0 ± 10.9	92.8 ± 2.7	–	89.2 ± 3.5	$\mathbf{99.2 \pm 3.4}$

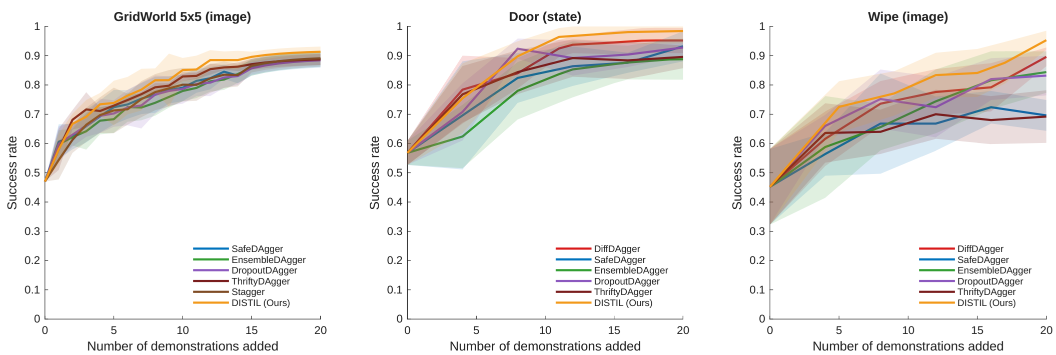

Figure 3: **Success rate vs. demonstrations added** on GridWorld 5×5 (image), Door (state), and Wipe (image); mean $\pm$ std over seeds (9 for GridWorld, 5 for the robot tasks). Under the same 20-demonstration budget, DISTIL tracks the pack over the early budget and separates over the second half, with the largest final margin on image-based Wipe, where the uncertainty-gated baselines plateau.

collected; the policy retrains from scratch after every demonstration on GridWorld and after every fourth demonstration on the robot tasks (the cadence of Lee, Kang, and Kuo 2025, shared by every method), and between retrains the policy is unchanged, so on the robot tasks up to three consecutive rounds analyze failures of the same policy. Held-out monitoring (the learning curves) uses a fixed set shared across methods and seeds: 200 GridWorld layouts, evaluated every round, and 100 episodes per robot task (frozen seeds), evaluated at every retraining checkpoint. Table 1 instead pools the evaluation pipeline's records at the budget, whose per-cell counts vary (caption). We run **9 seeds** per GridWorld cell and **5** per robot cell. DISTIL consumes inputs that cost no expert demonstrations (the authored knowledge graph, object poses for the state-run descriptors and for the memory centroids of every run, image runs included, and simulator resets for bridging starts); the baselines' gates receive free expert-action access instead, and the budget charges every method the same 20 recorded demonstrations. The primary metric is the final held-out success rate (SR) at the budget. As a first analysis metric we report the *per-demonstration information gain*: the current policy's per-step loss on each newly collected demonstration, before any retraining on it (on the robot tasks that policy can be up to three demonstrations stale). For DISTIL we also report the Pearson correlation between prescription confidence and the realized improvement ΔSR. ΔSR is the change in the policy's success rate on the round-level rollout evaluation (the 20-layout pool on GridWorld; the round's 60 rollout episodes on the robot tasks) across the retraining that incorporates the demonstration. The before evaluation is the pre-retrain rollout set whose failures the prescription targeted (on the robot tasks, the rollout episodes since the previous retraining, all under the one unchanged policy); the after evaluation is the following block's rollouts, so demonstrations incorporated at one retraining share that checkpoint's ΔSR. ΔSR is differenced per retraining, never accumulated since round 0. Because the before set is where the targeted failures were just observed, per-round values

Table 2: **Per-demonstration information gain**: the current policy's pre-retrain loss on each newly acquired demonstration, mean over each cell's 168–184 pooled loss records. A record is one logged evaluation of the policy's loss on a demonstration; on the robot tasks (5 seeds $\times$ 20 demonstrations) the analysis logs hold more than one record per demonstration, so the counts are records, not distinct demonstrations (per-record spreads for GridWorld image in Figure 4, top). Abbreviations as in Table 1; D-DAg is Diff-DAgger. DISTIL is the most informative in all ten settings.

Task	Obs.	Safe	Drop.	Ens.	Thr.	Stag.	D-DAg	DISTIL
GridWorld	state	2.55	2.95	1.33	1.34	1.83	–	**3.55**
GridWorld	image	2.46	2.53	1.57	1.37	1.88	–	**3.21**
Push-T	state	1.66	2.36	1.11	1.10	–	1.57	**2.81**
Push-T	image	2.04	2.16	1.06	1.20	–	1.80	**2.82**
Lift	state	2.23	2.10	1.12	1.13	–	1.61	**2.64**
Lift	image	2.18	2.17	1.00	1.21	–	1.36	**2.93**
Wipe	state	2.02	2.38	1.23	1.18	–	1.43	**2.91**
Wipe	image	2.50	2.96	1.43	1.52	–	1.95	**3.62**
Door	state	2.53	3.10	1.43	1.42	–	1.84	**3.43**
Door	image	2.35	2.46	1.24	1.26	–	1.58	**3.00**

run above the held-out curve's increments; and consecutive rounds evaluate different fresh draws, so their sum is not the held-out rise.

Main Results (Q1)

Table 1 reports final held-out success rates in all ten cells; Figure 3 shows learning curves on three representative ones. **DISTIL attains the highest mean success rate in every cell.** On state Push-T it reaches 96.1% against 90.7% for Diff-DAgger, a $+5.4$-point margin on the task family Diff-DAgger was designed for. Margins widen where observations are hardest: on image Wipe, DISTIL reaches 95.3% vs. 89.6% for Diff-DAgger and 69.6% for SafeDAgger ($+25.7$ points), consistent with a gate that keeps re-correcting the same contact-rich mode while the uncertainty-gated curves of Figure 3 flatten late in the budget. On image Door, DISTIL reaches 99.2%, $+6.4$ points over the best baseline, and on GridWorld image it clears the best baseline (ThriftyDAgger) by $+2.5$ points. Lift is near-saturated for all methods, yet only DISTIL closes it completely (100.0 ± 0.0 in both modalities). With 5–9 seeds per cell, the one-standard-deviation spreads of Table 1 overlap the best baseline in most cells, including Push-T and Wipe, so we claim no per-cell statistical separation. The evidence is the consistency of the ranking: the best mean in all ten cells, the largest margins on Push-T, Wipe, and image Door, and the best per-demonstration information gain in all ten cells (Table 2).

Information Gain and Its Allocation (Q2)

Table 2 and Figure 4 quantify how much new information each acquired demonstration carries: its pre-retrain policy loss (protocol section). A high pre-retrain loss admits three readings: (a) the demonstration covers a region the current policy has not learned; (b) the demonstration is suboptimal

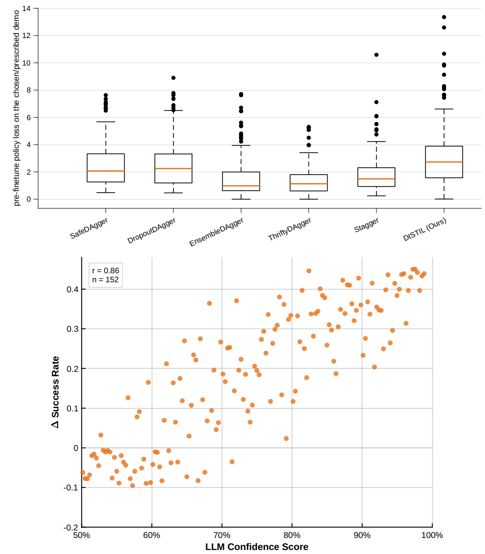

Figure 4: **Top: per-demonstration information gain** (GridWorld 5×5, image): the policy's pre-retrain loss on each newly acquired demonstration (the axis label *pre-finetune* names this pre-retrain loss; no fine-tuning is performed); DISTIL's prescribed demonstrations carry the most new information. **Bottom: prescription confidence predicts policy improvement** (same setting): each point is one LLM-prescribed demonstration ($r = 0.86$), with ΔSR the change in the round's rollout success rate across the retraining that incorporates it (protocol section); rounds resolved by the deterministic fallback emit no confidence and are excluded. Across all ten task–modality settings, Pearson r ranges from 0.82 to 0.89.

or invalid; (c) the region is represented but hard to fit (multimodal expert actions; noise in the stochastic denoising-loss estimate). The feasibility check and the expert largely rule out (b), leaving residual expert error (the Push-T PPO expert is strong but not provably optimal). Reading (c) is damped rather than excluded: the diffusion policies model multimodal actions by design (Chi et al. 2023), and the loss estimate averages over sampled noise and timesteps. The metric is therefore a novelty proxy, high when a demonstration comes from a poorly fit region, not a formal information measure. DISTIL's prescribed demonstrations have the highest mean gain in all ten cells (e.g., 3.62 vs. next-best 2.96 on image Wipe; 3.55 vs. 2.95 on state GridWorld), with the widest upper spread in the boxplot. Because DISTIL by construction seeks high-loss regions, its top rank here confirms that the mechanism operates as designed; it is not an independent quality measure. And gain alone does not predict final success: SafeDAgger and DropoutDAgger out-gain EnsembleDAgger and ThriftyDAgger, yet the gain orderings do not match the success-rate orderings of Table 1. What converts gain into success rate is its *allocation across failure modes*: a gate that repeatedly harvests high-loss demonstrations from the *same* mode accumulates redundant information; DISTIL's clustering and memory spread them over distinct, not-yet-corrected

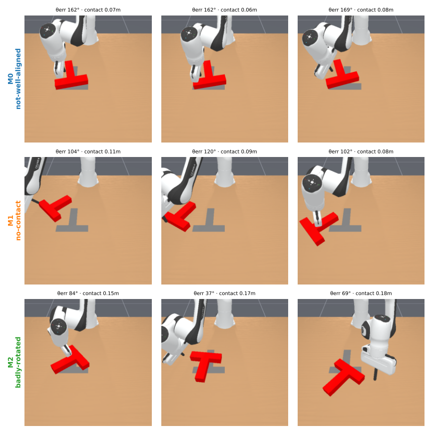

Figure 5: **Failure modes discovered on image-based Push-T**: k=3 recurring modes (*not-well-aligned*, *no-contact*, and *badly-rotated*), with three representative frames each, annotated with orientation error and contact distance. Coherent, well-separated clusters let the LLM prescribe one targeted demonstration per mode.

modes, a reading that matches the late-flattening baseline curves of Figure 3.

Confidence as an Improvement Predictor; Discovered Modes (Q3)

Each prescription carries a self-reported confidence emitted blind to the outcome: success-rate feedback arrives only after collection, retraining, and new rollouts. Figure 4 shows that this blind, self-reported confidence correlates with the realized policy improvement on image GridWorld ($r = 0.86$; rounds resolved by the deterministic fallback emit no confidence and do not enter the correlation, which excludes 28 of that cell's 180 rounds). Because ΔSR is differenced per retraining rather than accumulated over training, the correlation is not a byproduct of confidence and success both drifting upward with round index. The same computation per cell yields $r = 0.88$ (state GridWorld), $0.87/0.88$ (state/image Push-T), $0.88/0.89$ (Lift), $0.82/0.86$ (Wipe), and $0.83/0.82$ (Door), a 0.82–0.89 range across all ten settings. One caveat applies to the eight robot-task values: demonstrations incorporated at one retraining share that checkpoint's ΔSR, so those correlations effectively relate per-checkpoint confidence to per-checkpoint improvement, resting on about 25 outcome values per cell (5 seeds $\times$ 5 retrainings), not 100 independent ones. In the plotted GridWorld cell, low-confidence prescriptions ($< 60\%$) yield near-zero improvement while high-confidence ones ($\geq 90\%$) yield the largest gains, so a practitioner could veto low-confidence prescriptions before spending expert effort. Figure 5 shows the modes discovered on image-based Push-T: three coherent, well-separated clusters; these modes are the units over which the budget is allocated.

Conclusion

Interactive imitation learning has long asked *when* to query the expert. Under a fixed budget that is the wrong question: performance is set by *which* failures are corrected and *how* each demonstration is placed. DISTIL operationalizes this as demonstration distillation: locate each failure's peak-loss step with the learner's own loss, root-cause the failures with KAG-grounded foundation models, cluster them into modes under a cross-round memory, and prescribe each demonstration through a feasibility-checked LLM loop that corrects a failure in place or bridges several related ones. Across five tasks, two modalities, and four learner architectures, the same loop attains the best mean success rate against six interactive baselines (five per setting) at a 20-demonstration budget, and its demonstrations carry the most new information, spread across distinct failure modes.

Limitations. Our robot-task experts are oracles (a privileged PPO policy and scripted motion planners); noisy human demonstrators would add variance these oracles do not capture (Mandlekar et al. 2020; Liu et al. 2023). Bridging placements require a resettable simulator, and all results are in simulation. The robot-task comparison lacks a uniform-random allocation control (Stagger runs on GridWorld only). The loop inherits the quality of its foundation models: weaker VLM or LLM outputs degrade root-causing and prescription; KAG grounding and the feasibility loop bound the damage without eliminating it. Each round also pays the wall-clock cost of the reasoning pipeline, which per-state gates do not incur, and each new task family requires authoring its knowledge graph. The loop commits to one demonstration per round, and the descriptor ϕ_i and memory constants are designed, not learned. Letting the VLM's failure taxonomy define the clustering space and extending prescription to human-gated hardware (Kelly et al. 2019) are natural next steps.

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